Week : 12 (16-20 November 2026)

الأسبوع #: من - إلى

Subject/الموضوع: Advanced Physics **Unit Title عنوان الوحدة: Torque** **Grade الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Scientific Poster – Diagrams & Calculations الحصة الأولى	• I can finalise my labelled torque diagram and calculations for my balanced structure, using $\tau = rF \sin\theta$ to prove that $\Sigma\tau = 0$.	3.3 Guide processing with explicit steps, chunking, and interactive models. 6.2 Support planning with templates, strategy prompts, and modeled thinking.	GRASP instructions, rubric and poster template on Toddle; calculator; physics notebook.
Lesson 2 Scientific Poster – Evaluation & Final Edits الحصة الثانية	• I can explain how my design relates to safe UAE infrastructure, evaluate its limitations and finalise my poster against the GRASP rubric.	5.2 Use multiple tools for composition (spellcheckers, manipulatives, templates). 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	GRASP rubric on Toddle; digital poster software; physics notebook.
Lesson 3 GRASP Presentations (Part 1) الحصة الثالثة	• I can present my balanced structure and scientific poster, explaining the pivot, forces, lever arms and equilibrium proof, and answer questions using physics evidence.	5.1 Provide multiple communication media (text, speech, visuals, models). 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement.	Completed poster and physical model; GRASP rubric on Toddle; projector/screen; peer feedback form.
Lesson 4 GRASP Presentations (Part 2) & Reflection الحصة الرابعة	• I can justify my design with calculations during my presentation, give constructive peer feedback and reflect on my learning against the GRASP rubric.	5.1 Provide multiple communication media (text, speech, visuals, models). 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Completed poster and physical model; GRASP rubric and submission instructions on Toddle; peer feedback and self-reflection forms.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Lessons 1–2 take place in class: students finalise their scientific poster (labelled diagram, data table, torque calculations, proof that $\Sigma\tau = 0$ and UAE connection). Lessons 3–4: GRASP presentations – students bring their completed physical model and present their poster. The presentation order is shared on Toddle and the audience completes peer feedback. Final poster is submitted on Toddle.		

Week #12 الأسبوع#

Technology Integration دمج التكنولوجيا	Digital poster design or presentation software; calculator; projector/screen for presentations. • Toddle for the GRASP brief, rubric, peer feedback and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: trigonometry, moments and equations. Engineering: stability, structures and safety. Design: scientific communication. • Engineering: rotating shafts and wheels in machines. • Design & Technology: gears, pulleys, rotational systems.
External Assessment Integration دمج التقييمات الخارجية	AP Physics-style mathematical reasoning, force diagrams, units and evidence-based justification.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Critical thinking through solving equilibrium problems. • Transfer skills: linking linear and rotational analogies. • **Research Skills:** • Investigating moments of inertia for real objects. • Using simulations and analyzing data. • **Communication Skills:** • Explaining diagrams, torque direction, and reasoning. • Collaborative problem-solving dialogue in groups. • **Wellbeing:** • Promotes confidence through hands-on activities and manipulatives. • Reduces anxiety by chunking complex rotational concepts into simpler steps. • Encourages peer support in groups.
Standards المعايير	AP Physics 1 – Unit 5: Torque and Rotational Dynamics • Topic 5.1 – Rotational Kinematics: Describe rotational motion using angular quantities. • Topic 5.2 – Connecting Linear and Rotational Motion: Relate force, distance and angle to rotational effects. • Topic 5.3 – Torque: Calculate torque and determine its clockwise or counterclockwise direction. • Topic 5.5 – Rotational Equilibrium: Apply the condition $\Sigma\tau = 0$ to analyse balanced systems and justify equilibrium.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can engineers prove that a UAE-inspired structure will remain balanced and safe?

Success Criteria معايير النجاح (Measurable)	**Lesson 1 – Scientific Poster: Diagrams & Calculations** MUST (DOK 1–2): Complete a labelled diagram showing the pivot, at least two forces, lever arms, angles and torque directions. SHOULD (DOK 3): Calculate each torque using $\tau = rF\sin\theta$, with correct units and directions, and show mathematically that $\Sigma\tau = 0$. COULD (DOK 4): Evaluate how changes in force, angle or distance would affect equilibrium and recommend corrections. **Lesson 2 – Scientific Poster: Evaluation & Final Edits** MUST (DOK 1–2): Complete the poster with a labelled diagram, data table, equations and UAE connection. SHOULD (DOK 3): Explain and justify the equilibrium proof and its relevance to safe UAE infrastructure. COULD (DOK 4): Evaluate design limitations and propose evidence-based improvements. **Lesson 3 – GRASP Presentations (Part 1)** MUST (DOK 1–2): Present the structure, poster and equilibrium proof clearly, using correct physics vocabulary and units. SHOULD (DOK 3): Justify design choices with torque calculations and answer questions using evidence. COULD (DOK 4): Compare the design with real UAE structures and defend proposed improvements. **Lesson 4 – GRASP Presentations (Part 2) & Reflection** MUST (DOK 1–2): Present remaining designs clearly and complete the peer feedback form. SHOULD (DOK 3): Give specific, evidence-based peer feedback linked to the GRASP rubric. COULD (DOK 4): Reflect critically on own design and presentation, identifying strengths and next steps.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging: responsible engineering design. Global Understanding: safe, sustainable infrastructure.
	Area: Innovation and Infrastructure
	Students connect rotational equilibrium to UAE cranes, traffic signs, stadium lighting, solar panel arms and other engineered structures. Their scientific posters and presentations explain how balanced designs contribute to public safety, innovation and national development. This topic directly supports the **UAE National Identity Program: Science, Technology, and Engineering (Innovation & Infrastructure domain)** by developing students' ability to apply physics principles—such as torque and rotational equilibrium—to real-world engineering systems. Understanding how forces create balance and stability underpins safe infrastructure design, mechanical efficiency, and problem-solving skills essential for the UAE's Vision 2031 focus on innovation, sustainable development, and future-ready engineers.